An extensive dataset of spiking activity to reveal the syntax of the ventral stream

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Summary

Visual neuroscience benefits from high-quality datasets with neuronal responses to many images. Several neuroimaging datasets were published in recent years, but no comparable dataset with spiking activity exists. Here, we introduce the THINGS Ventral-stream Spiking Dataset (TVSD). We extensively sampled neuronal activity in response to >25,000 natural images from the THINGS database in macaques, using high-channel-count implants in three key cortical regions: V1, V4, and the inferotemporal cortex. We showcase the utility of TVSD by using an artificial neural network to visualize the tuning of neurons. We also characterize the correlated fluctuations in activity within and between areas and demonstrate that these noise correlations are strongest between neurons with similar tuning. The TVSD allows researchers to answer many questions about neuronal tuning, the interactions within and between cortical regions and to compare spiking activity in monkeys to human neuroimaging data.

Keywords

Open Data, Vision, Spiking activity, Ventral stream, Natural images

Introduction

Making sense of a visual scene is a complex and computationally intensive process, and its neuronal foundations in our brains remain far from being understood. Neuroscientists have traditionally approached this problem by investigating how neurons in various cortical regions of the ventral visual stream, involved in object recognition, respond to features of the visual world (e.g. color and orientation), using controlled stimuli[1,2]. More recently, however, scientists have also started to incorporate novel, data-driven approaches to capture neuronal representations of the ventral stream, measuring brain responses to large collections of ecological stimuli[3]. This approach gained popularity due to the astounding performance of computational models, developed in the artificial intelligence (AI) field, in predicting brain activity[4–7]. These new computational models depend on high-quality and large-scale datasets with neuronal responses to thousands of images[2,8–12]. In recent years, several large-scale neuroimaging datasets have been recorded and publicly shared, allowing researchers around the world to address their research questions using data of high quality[8–11].

The existing neuroimaging datasets offer extensive brain coverage and stimulus variability, but to deepen our understanding of the syntax of the ventral stream, it is useful to investigate the spiking activity and tuning of neurons using intracortical electrodes, which provide a higher spatiotemporal resolution than neuroimaging techniques. Such a dataset does not yet exist, presumably because it is challenging to sample neuronal responses across enough images and neurons, in multiple brain regions at once, across multiple days. Here we overcame these challenges by chronically placing many Utah arrays in key regions of the ventral stream using custom-made implants.

We present the THINGS Ventral stream Spiking Dataset (TVSD), based on large-scale electrophysiology experiments, which provides public access to spiking responses from visual neurons elicited by many images. The THINGS database is an extensive and carefully curated set of naturalistic images, for which human judgments of category membership have been analyzed systematically[12]. We sampled the multi-unit spiking activity (MUA) in two macaques seeing more than 25,000 natural images from the THINGS database using high-channel-count implants with 1,024 microelectrodes in three key regions of the ventral stream: the primary visual cortex (V1), area V4, and the inferotemporal cortex (IT). We provide the raw data sampled at 30kHz and the pre-processed MUA, which is easy to use by researchers with no prior knowledge of intracortical recordings. To demonstrate the potential of the data set, we showcase a modeling approach based on the pre-processed MUA and analyze the patterns of correlated fluctuations in the firing rates across V1, V4 and IT.

We expect that the TVSD will allow scientists to test theories on the mechanisms for natural vision, going beyond what has been possible with the existing datasets. Because THINGS was used in several human behavioral and neuroimaging public datasets, the TVSD will enable unprecedented comparisons between spiking data obtained in monkeys and neuroimaging and behavioral data in humans. Finally, the size of the TVSD could unlock the possibility of training AI models directly on brain data. The full dataset is available on G-Node GIN at the link: https://doi.org/10.12751/g-node.hc7zlv.

Results

High-channel-count neuronal responses in the ventral stream to thousands of images

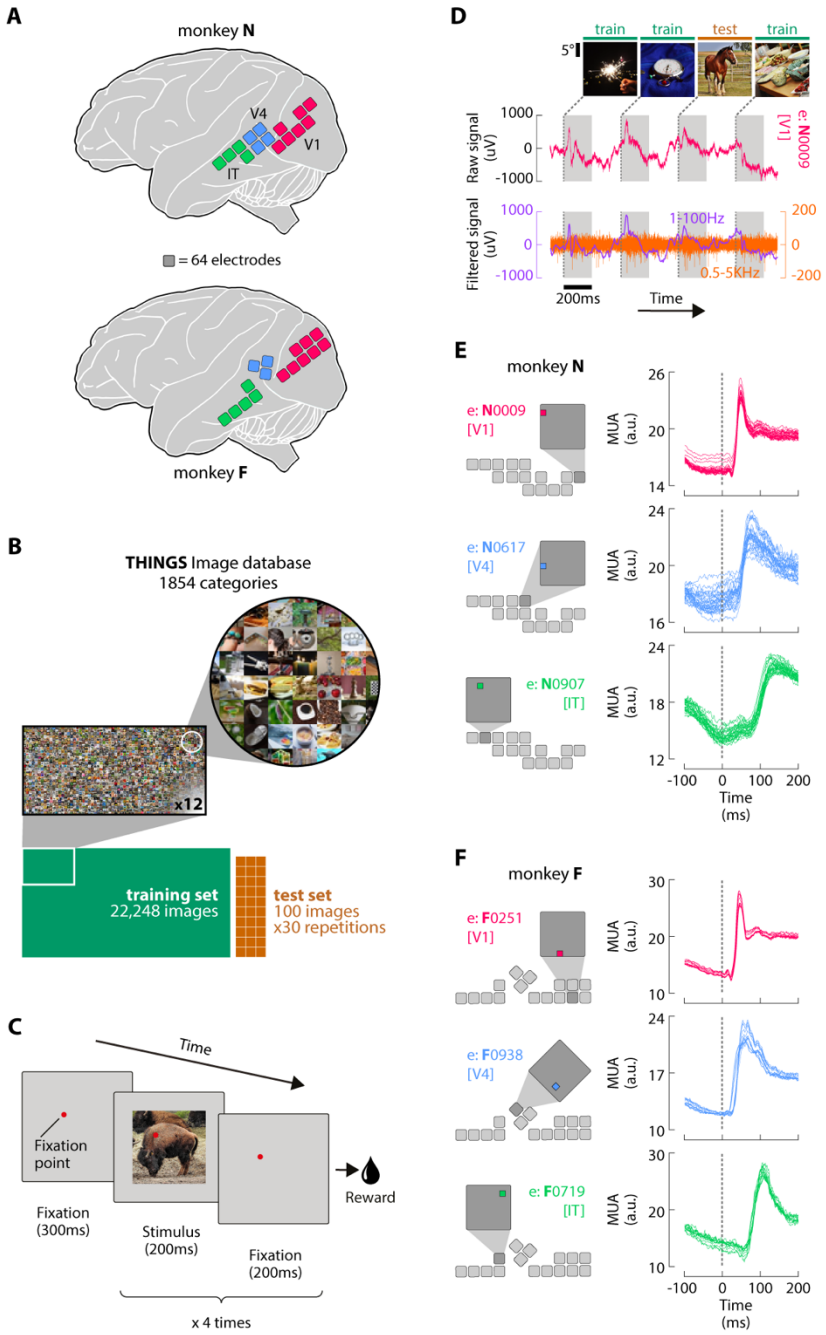
We implanted 15 or 16 Utah arrays in two macaque monkeys[13,14]. Each array had eight rows and columns of 1.5-mm-long shanks (Figure 1A, Figure S1). We implanted a total of 15 arrays in V1 (7 in monkey N and 8 in monkey F; 960 electrodes in total), 7 arrays in V4 (4 in monkey N and 3 in monkey F; 448 electrodes in total) and 9 arrays in IT (4 in monkey N and 5 in monkey F; 576 electrodes in total).

We presented 25,248 natural images from the THINGS database[12] ensuring that they covered the receptive fields (RFs) of the neurons, during a task in which the monkeys directed their gaze to a fixation point. The images were divided into a training and a test set, but their presentation was randomized and interleaved. The training set spanned 1,854 systematically curated semantic categories, with 12 images per category, which were shown once. The training set will allow users to study the neuronal responses to a wide range of stimuli and compare them to those predicted by models of cortical processing, including deep neural networks. The test set comprised 100 images that were shown 30 times each (Figure 1B). The purpose of the test set was i) to compare responses elicited by the same images across days[7] (e.g. it is possible to normalize responses to these images across sessions), ii) to increase the signal-to-noise ratio (SNR) and iii) to enable cross-validation[1].

A trial started with the presentation of a fixation point, a red dot on a uniform gray background. Once the monkeys attained fixation for 300ms, we presented a sequence of four images, presented for 200ms and followed by 200ms with a gray screen (Figure 1C). The monkeys were rewarded with a drop of juice if they maintained fixation during the whole sequence. We recorded four sessions in each monkey. The sessions spanned 7 days in monkey F with 6,952 trials and 25,248 image presentations, and 4 consecutive days in monkey N with 6,771 trials and 25,248 image presentations. Figure 1D shows the raw electrophysiological

signal, sampled at 30kHz, from an example trial at a representative electrode in V1 from monkey N. For our analysis we will here focus on MUA, which represents the spiking activity of several neurons in the vicinity of the electrode (Methods).

MUA population responses are very similar to those obtained by pooling across single units and have been used in many previous studies[15–18]. Trautman et al. (ref.[19]) demonstrated that neuronal dynamics are similar when analysing multiunit data and single neurons, and that similar scientific conclusions can be reached about population-level activity. The MUA was stable within and across the recording sessions, because the average activity in V1, V4 and IT elicited by all images in blocks of ~20 min across all sessions was similar (Figure 1E,F). Nevertheless, users of the TVSD can apply other filters to e.g. extract the local field potential from the broad-band signal. They can also use spike-sorting algorithms isolate single units (examples of spikes sorted with Waveclus3[20] are presented in Figure S2).



<u>Figure 1. Spiking activity in the monkey ventral stream in response to thousands of images</u>
A) Schematic of the array locations. We recorded spiking activity from 31 Utah arrays (8 x 8 electrodes each) implanted in two macaque monkeys. Figure S1 shows pictures that were taken during surgery and can be used for the precise determination of array locations.
B) We presented 25,248 images from the THINGS image database[12], which were divided into a training and a test set. The rectangle encloses 1,854 images, which corresponds to 1/12th of the full training set.
C) The monkeys maintained fixation during a sequence of four images (200ms on, 200ms off). Successful fixation during the entire sequence was followed by the delivery of a juice reward.

D) Broad band signal at a representative electrode (N0009) in V1 during an example trial (top) and the lowpass LFP and bandpass spiking components (bottom; purple: 1-100Hz; orange: 0.5-5KHz). The raw signal contains high frequency spiking activity and slower LFP components. Dashed line: stimulus onset; shaded gray region: stimulus ON period.
E-F) MUA at representative electrodes in monkey N (E) and monkey F (F), averaged across all the trials in blocks of ~20min. From the top: a V1 electrode (red), V4 electrode (blue) and IT electrode (green). The electrode position in the array is shown on the left. The small variability across response averages illustrates the stability of the signal within and across sessions. Figure S2 shows example results from a spike-sorting algorithm.

Quality, reliability and selectivity of spiking activity in V1, V4 and IT

We assessed the quality of the recordings by measuring the SNR and the reliability of the MUA at individual recording sites. We measured the SNR by dividing the peak of the average response elicited by all images by the standard deviation of the activity before the stimulus onset (see Methods). Natural images have a lower contrast and higher variance than the artificial stimuli of more standard experiments (e.g. checkerboards) and some of the images elicited inhibition. The SNR is therefore a conservative measure, but it exceeded 2, a conventional threshold for electrode inclusion[15–17], for most recording sites (Figure 2A; 74.3% in monkey N; 54.0% in monkey F).

The SNR of the TVSD is considerably higher than state-of-the-art neuroimaging datasets, such as the Natural Scenes Dataset (NSD[8]). The authors of the NSD computed SNR based on the variance of responses elicited by the same images ('*ncsnr*'[8]), and this SNR measure is 6 times larger for the TVSD. They proposed the concept of 'equivalent trials', which is proportional to the number of trials and the square of the SNR of the data and can be used to compare datasets. The authors noted that the NSD expanded the equivalent trial number by a factor of ~22 over previous work. According to this calculation, the TVSD would increase the number of equivalent trials by an additional factor of 10.

To measure the reliability of the neuronal responses, we used the oracle correlation[5], which is the correlation between the average response elicited by 29 of the 30 test stimuli with the response to the one stimulus repetition that was held out (see Methods). Most channels had a significant oracle correlation ($p < 0.05$, t-test, Bonferroni corrected): 87.3% in monkey N and 75.0% in monkey F (Figure 2B). The oracle correlation can be interpreted as the noise-ceiling, i.e. the maximum correlation of a model accurately predicting the neuronal responses elicited by individual images. Interestingly, many channels with low SNRs had a high oracle correlation (e.g. the example recording site on the right in Figure 3A). This apparent discrepancy is caused by the definition of SNR, which was based on the average response and did not take the neuronal tuning or the presence of inhibitory responses (Figure 3A,C) into account. We therefore also computed another measure of the signal-to-noise ratio, SNR_{Max}, which is the response to the preferred test image, normalized to the standard deviation of activity before stimulus onset (Figure S3A).

We next examined the latencies of the responses (Figure 2C). As expected, V1 neurons were activated first, at a mean latency of 32ms in both monkeys. This was followed by area V4 with latencies of 38 and 51ms, and then by IT with latencies of 58 and 66ms in monkeys N and F, respectively.

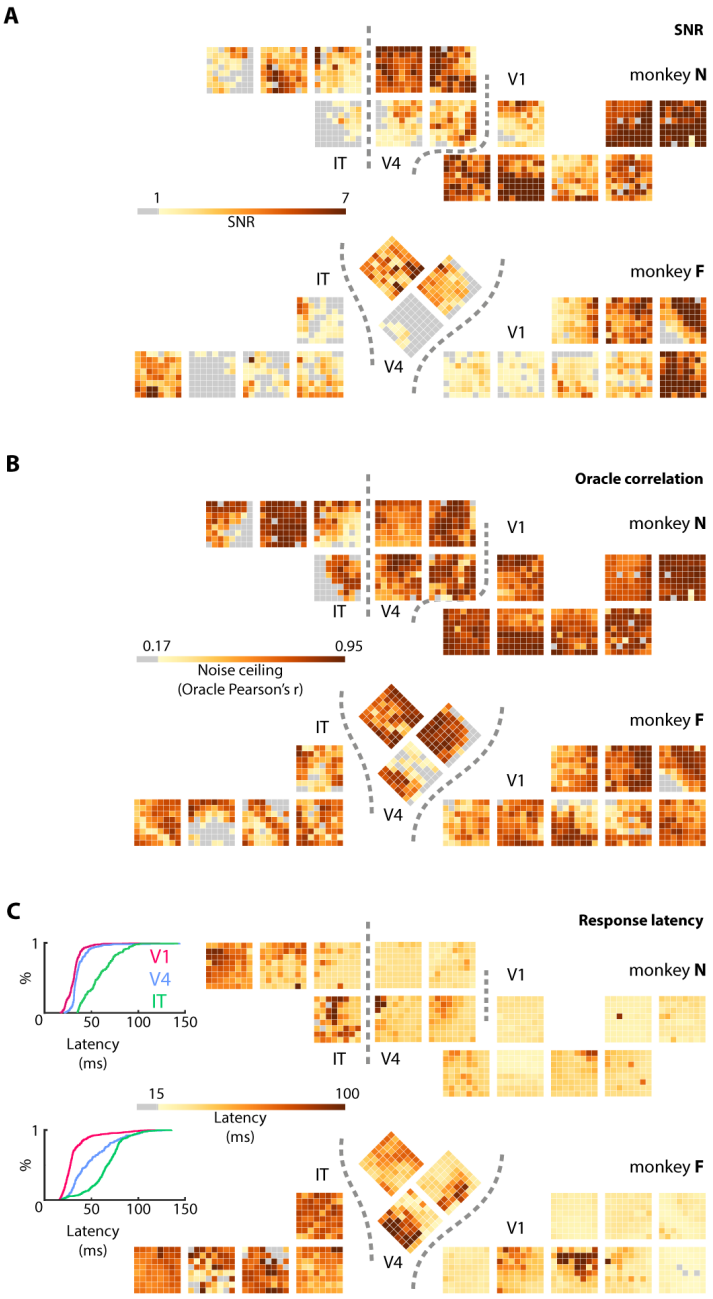


Figure 2. Signal-to-noise ratio, reliability and latency of neuronal responses in V1, V4 and IT
A) The SNR at each recording site. Electrodes with SNR < 1 are shown in gray. Figure S3A shows SNR_{Max}, computed for the test stimulus that elicited the strongest activity at a recording site.
B) The oracle correlation as measure for the noise ceiling and the reliability of the responses. Most recording sites had a significant oracle correlation (gray, non-significant oracle correlations).
C) The latency of the neuronal responses was shortest in V1, longer in V4 and even longer in IT (gray, unreliable estimates). Insets on the left show the cumulative distributions of latency across recording sites in V1 (red), V4 (blue) and IT (green). Figure S3B shows the same distributions as histograms.

We examined the neuronal selectivity to semantic classes (Figure 3). For each electrode we determined which of the 1,854 image categories evoked the highest response and assigned the preferred category to a supra-ordinate semantic class, which had been estimated in a previous behavioral study with the THINGS images[21]. Figure 3A shows the stimulus selectivity of two representative recording sites in IT, one from each monkey. Each dot represents the normalized response to a single image repetition from the test set (100 images, 30 repetitions each), in the time window 75-175ms, centered around the peak of response. We normalized the responses by subtracting the mean and dividing by the standard deviation of the responses to the images in the test set shown in the same session (Methods) and sorted them according to the mean response. For visualization purposes we focused on the test images, because using all images of the training set gave rise to a crowded plot. Both sites were highly selective for stimuli containing animals, with pictures of monkeys eliciting the highest responses and pictures of inanimate objects eliciting the lowest responses (Figure 3A).

Figure 3B shows the recording sites where one of the top-5 classes elicited the strongest response in the training set (top: monkey N; bottom: monkey F). In both monkeys, many IT recording sites were tuned to animals (47% in monkey N; 21% in monkey F), food (7% in monkey N; 13% in monkey F) or body parts (6% in monkey N; 4% in monkey F)[22–26]. Figure 3C, shows the highest and lowest responses elicited by the test images, averaged across recording sites in each region. The responses had a wide range, and some of the stimuli elicited inhibitory responses. We also obtained selective responses in V1 and V4, but it is likely that this selectivity does not reflect tuning to semantic classes, but to the local features in the RFs that are more frequent in images of specific classes. Users of the TVSD can examine the

tuning in V1 and V4 by extracting the image features that fell in the RFs of the V1 and V4 neurons (see below for an analysis of RFs).

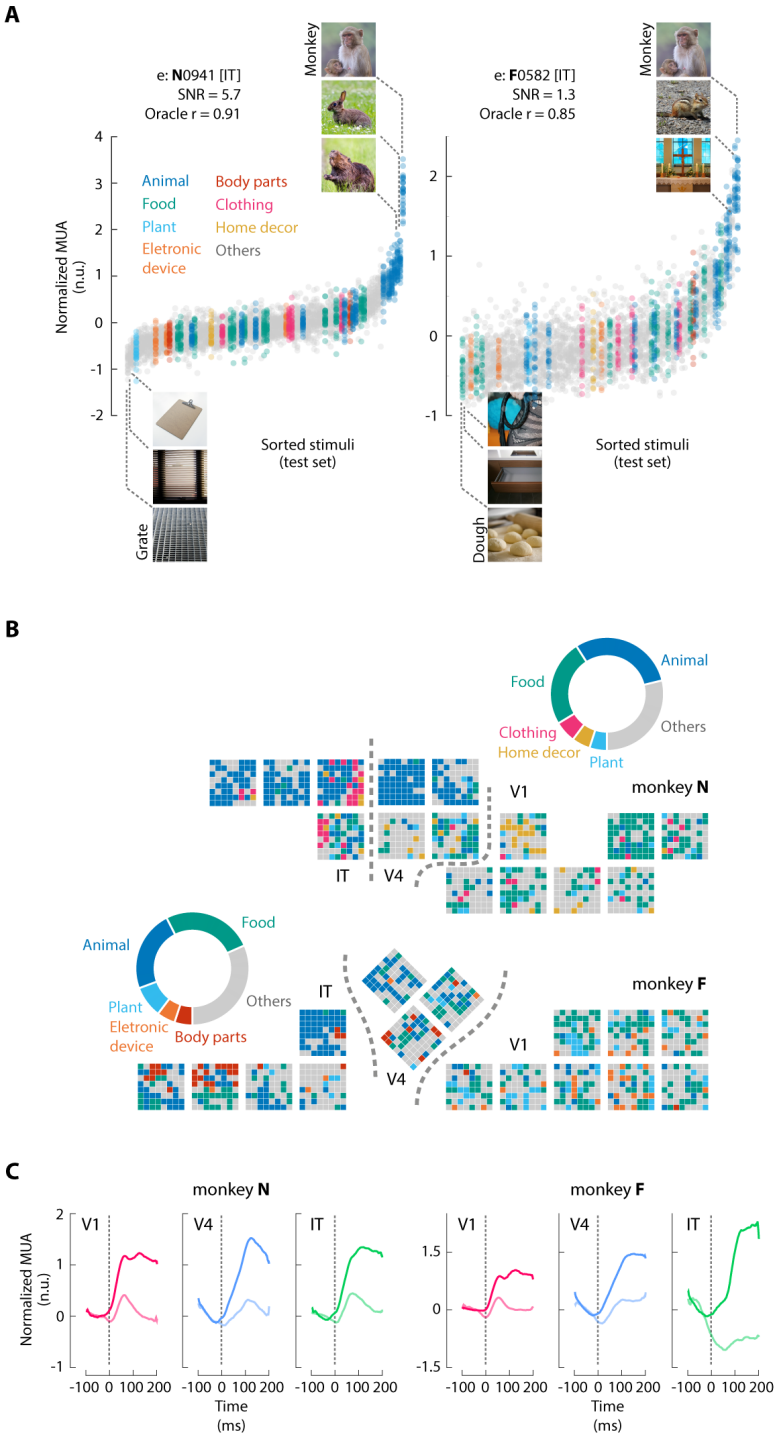


Figure 3. Selectivity for semantic classes of THINGS images

A) The selectivity of example IT recording sites in monkey N (left) and monkey F (right). Data points represent the normalized MUA response in a time window from 75-175ms to single

image presentations (N=30 repetitions per image). Insets, the three images eliciting the highest and lowest responses, which suppressed activity. Dot colors correspond to the supra-ordinate semantic classes of the images (the images shown are license-free and differ from those in the test set). The colors represent top 5 semantic classes, which differed between the two monkeys. For example, body parts belonged to the top 5 classes for monkey F but not for monkey N.
B) Selectivity for the semantic classes of THINGS images. The colors illustrate the supra-ordinate semantic class[21] of the objects eliciting the strongest responses (top: monkey N; bottom: monkey F).
C) Highest and lowest responses in each region for each monkey to the test images, averaged across repetitions and recording sites (using a 25ms smoothing kernel; V1: red; V4: blue; IT: green; dashed line indicates the onset of the visual stimulus).

Showcase #1: AI modeling of the response and the most exciting input at individual recording sites

To illustrate the potential of the TVSD for AI applications, we modeled the MUA responses at each recording site with the approach of Bashivan et al.[7] (including recording sites with a mean reliability higher than 0.3, see Methods; monkey N: N=759; monkey F: N=606). For this analysis, we used the normalized responses, averaged in a time window centered on the peak response, which differed across the areas (V1: 25-125ms; V4: 50-150ms; IT: 75-175ms). We determined the linear projection from the layer of a convolutional neural network (CNN, Inception v1[27], e.g. layer 'conv2d2' or 'mixed4e') that best predicted the response of neurons at each of the recording sites. This prediction also yielded an estimate of the RF of neurons at each of the recording sites (Figure 4) (details in Methods).

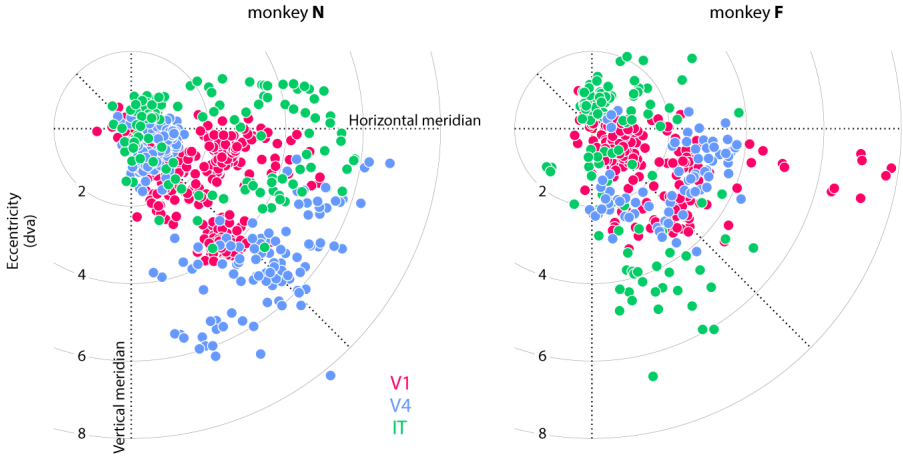


Figure 4. RF centers of recording sites.
RF centers of recording sites in monkey N (left) and monkey F (right), estimated by the model that predicts responses to the TVSD. Red, V1 sites; Blue, V4 sites; Green, IT sites.

To evaluate the quality of the model estimate of tuning, we measured the cross-validated correlation of the response predicted by each CNN layer (trained using only the images from the training set: see Methods) for the 100 test images in the dataset (Figure 5A). The model performance was high in each of the areas. The mean correlations were 0.75 and 0.76 in V1, 0.76 and 0.74 in V4 and 0.79 and 0.72 in IT in monkeys N and F, respectively. The inception layer number that was able to best model the tuning increased from V1 to V4 and was highest in IT (Figure 5B).

We used the model to estimate the most exciting inputs (MEIs)[5], which are images that are predicted to elicit a strong response and that illustrate the neurons' selectivity. Figure 5C shows a few representative MEIs from neurons at recording sites in the different areas in the two monkeys. The MEIs included edges and colorful patches in V1, more complex and texturized/curvilinear patterns in V4 that sometimes resembled eyes and furry animal faces or body parts at the example IT recording sites. We stress that these MEIs are based on MUA and represent images that elicit strong responses in groups of neurons and do not account for forms of tuning that differ between neighboring neurons. In area V1, for example, the

precise location of on- and off RF regions of neighboring simple cells might differ and would not be picked up by the model.

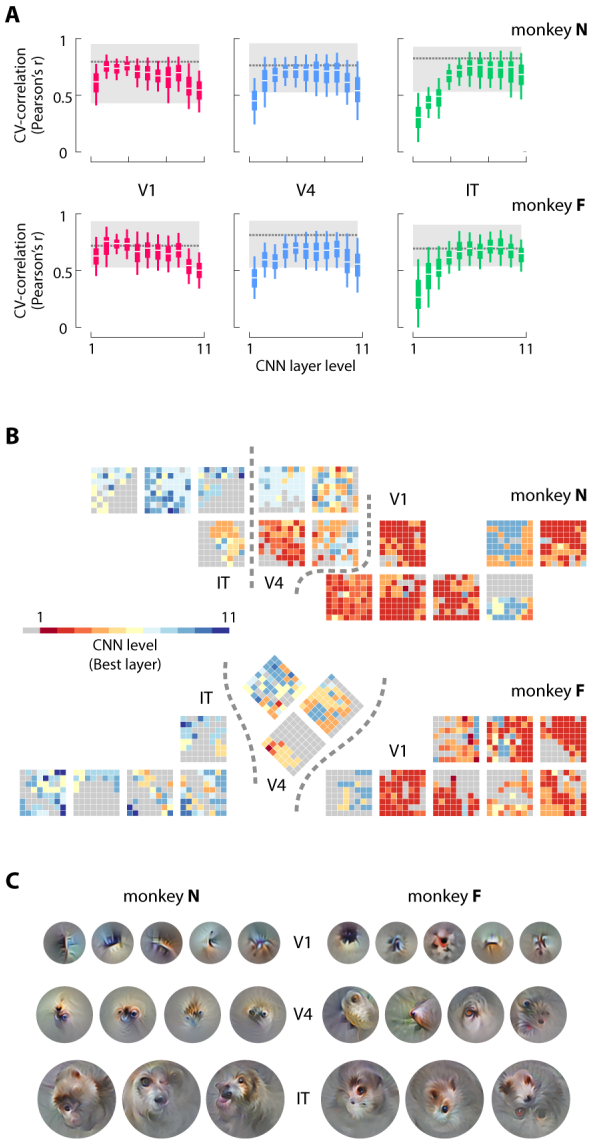


Figure 5. Modeling the tuning and the most exciting input of recording sites
A) Average cross-validated (CV) correlation based on 11 layers of the inception network in V1 (red), V4 (blue) and IT (green). Top, monkey N. Bottom, monkey F. White line, median. Box: 25-75th percentiles. Whiskers, range of all values. Dashed horizontal line, median oracle correlation across electrodes, which represents the noise ceiling. Shaded gray region, range of the oracle correlation across electrodes (i.e. min and max).
B) Inception layer with the highest CV-correlation for each recording site.
C) Representative MEIs from both monkeys in V1 (top), V4 (middle) and IT (bottom).

Showcase #2: Noise correlations within and between cortical areas

We analyzed the pattern of correlated fluctuations in activity across trials, known as noise correlations, as a second illustration of the utility of the TVSD. Noise correlations represent the joint fluctuations of activity across recording sites. They occur during the neuronal responses to specific pictures and during spontaneous activity[28–30]. Previous studies demonstrated that noise correlations are stronger between neurons that are tuned to similar features[31]. In our analysis, we split the 30 trials with each test stimulus into two halves. We used the odd-numbered trials to determine the noise correlations, and the even-numbered trials used to assess similarity of tuning of the neurons.

We illustrate the noise correlation for example pair of V1 recording sites in Figure 6A,B. The activity in a time-window from 25-125ms was correlated across the odd numbered trials, with a correlation coefficient of 0.78 (N=15 trials, $p < 10^{-6}$, t-test). The responses to the other test stimuli were also correlated across trials and we computed the pooled noise correlation, by first subtracting the average response per test stimulus and then correlating the single trial fluctuations around these averages (N=1,500 odd-numbered trials, Figure 6C, left)[32]. The pooled noise correlation for this example pair was 0.34 ($p < 10^{-6}$, t-test).

To measure of the similarity of the tuning of neurons at the two recording sites across test images, we computed the signal correlation, i.e. the correlation between the average responses elicited by the 100 test images[32]. A high signal correlation implies that neurons at the two sites prefer the same images. The example V1-V1 pair had a signal correlation 0.83 ($p < 10^{-6}$, t-test; Figure 6C, right). Figure 6D illustrates a weaker noise correlation for another pair of V1 sites ($p < 10^{-6}$, t-test) with RFs at a larger distance and with a weaker signal correlation.

The signal correlation is a straightforward measure of tuning similarity, but it depends on the stimulus set, which imposes a pattern of spatial and feature correlations across the

images. In natural images, nearby regions are generally more similar than regions that are farther apart, causing stronger signal correlations of neurons with nearby RFs[33–36]. Users of the TVSD can examine other measures such as similarity of tuning to color or orientation, as was mentioned in relation to showcase #1. The signal and noise correlation relate the activity at two recording sites but are mathematically independent. Neurons may prefer similar images but have independent activity fluctuations across stimulus repetitions (e.g. neurons in different individuals).

We next investigated the pattern of noise correlations across all V1-V1 pairs in monkey N (N=72,711). The median noise correlation was 0.11 and the median signal correlation 0.35 (Table 1; red bars in Fig. 7A). The noise correlation was weaker for recording sites with RFs that were farther apart (Fig. 6E) ($p < 10^{-6}$, t-test) and the signal correlation was weaker too (Fig. 6F) ($r=-0.59$; $p < 10^{-6}$, t-test). The strength of the noise correlation was predicted by the signal correlation, with a correlation coefficient of 0.28 ($p < 10^{-6}$, t-test) (Figure 6G), accounting for ~7% of the variance (R^2, square of the correlation coefficient). Hence, the RF-distance and signal correlation both predicted the noise correlation, and these predictors were also correlated among themselves. We used a variance partitioning analysis[25,37–40] to determine how much variance of the noise correlation was uniquely explained by the RF-distance, the signal correlation and how much variance could be explained by both factors (Fig. 7C). Of the variance, 3.4% was associated with both predictors, and the signal correlation explained 4.2% unique variance. In contrast, the RF-distance accounted for only 0.1% unique variance, indicating most of the variance associated with the RF-distance was also accounted for by the signal correlation. We also examined the noise correlations during the spontaneous activity (100ms before stimulus onset; Figure 7C, Table 1 and Figure S4) and found that they were stronger than during the visual stimulus ($p < 10^{-6}$, t-test). The signal correlation explained

16.1% of the variance in the spontaneous noise correlation ($p < 10^{-6}$, t-test). It is remarkable that signal correlations, induced by viewing natural images, predicted the spontaneous V1 activity fluctuations because they occurred in the absence of a visual stimulus. We replicated these results in monkey F, in which the median noise correlation during picture presentation was 0.07, the spontaneous noise correlation was 0.11 and the signal correlation 0.19 (N=41,905) (Table 1, Figure 7B,C).

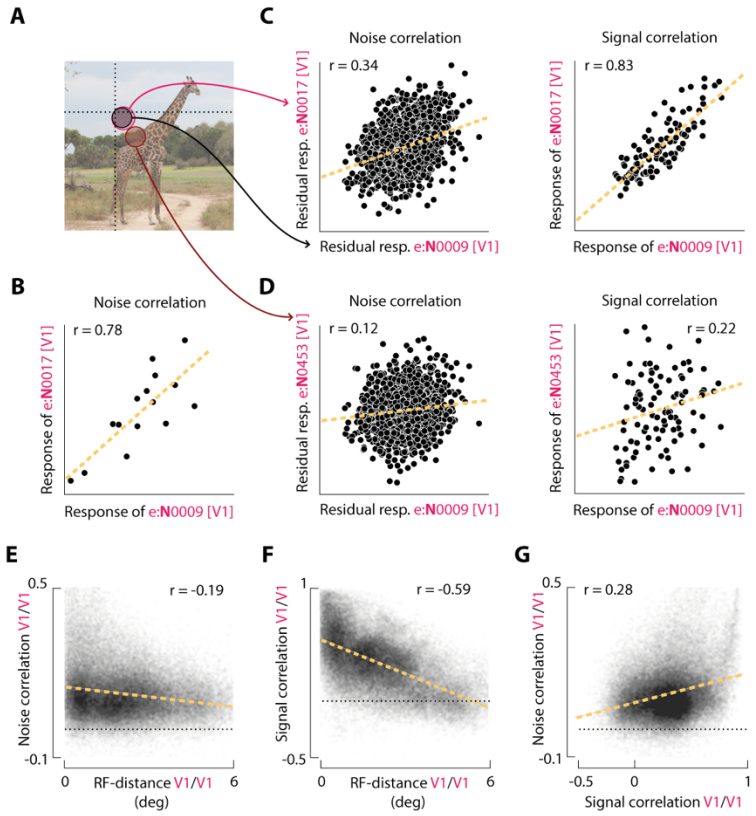


Figure 6. Noise correlations and signal correlations in V1.
A) Example test image with the RFs of three V1 recording sites (black, red and brown circles).
B) Correlation between the response of two V1 recording sites (black and red RFs in A) across 15 repetitions of the test image (25-125ms after stimulus onset).
C) Left: noise correlation of the same pair of V1 sites across 1,500 trials with the 100 test images, after subtracting the average response per image. Right: Signal correlation of the same pair of V1 sites, between the average responses to the 100 test images.
D) Same as C for another V1-V1 pair (red and brown RFs in A) recorded simultaneously.
E) Correlation between RF-distance and noise correlation across V1-V1 pairs in monkey N (N=72,711).
F) Correlation between RF-distance and signal correlation.

G) Correlation between the signal and noise correlation across V1-V1 pairs.

We also examined the correlations between neurons within V4 and within IT (Fig. 7A; Table 1). These correlation patterns bore similarities to those within V1. The median V4-V4 noise correlations during stimulus presentation (time-window centered on the peak response, 50-150ms) were 0.10 (N = 19,503) and 0.07 (N = 6,786), for monkeys N and F, respectively, and the median IT-IT noise correlations (time-window 75-175ms) were 0.09 (N = 9,870) and 0.08 (N =12,246). The signal correlations accounted for 25.4% and 20.0% of the variance in the noise correlations of V4-V4 pairs in monkeys N and F, respectively, and 42.7% and 10.5% of the noise correlations between IT-IT pairs. The spontaneous noise correlations within V4 and within IT were of comparable strength as during the visual stimulus (Table 1), and most of the variance was explained by the signal correlations (Figure 7C). Hence, the signal correlations induced by natural images are mirrored by activity fluctuations within V4 and IT in the absence of a stimulus, just as was the case for area V1.

We next investigated the noise correlations between neurons in different areas. Recent studies reported that noise correlations between V1 and V4 are generally positive[41,42], but the noise correlations between IT and lower visual areas have, to our knowledge, not been described. Figure 7D-E illustrates the noise correlations between example recording sites in V1, V4 and IT in monkey N. The noise correlation was 0.33 for the V1-V4 pair ($p < 10^{-6}$, t-test), 0.29 for the V1-IT pair ($p < 10^{-6}$, t-test) and 0.49 for the V4-IT pair ($p < 10^{-6}$, t-test) and the respective signal correlations were 0.42, 0.08 and 0.06. This example is remarkable, because it illustrates the coupling of activity fluctuations of neurons in V1 and IT, far apart in the visual cortical hierarchy.

Table 1. Medians of the noise and signal correlations.

	V1/V1	V4/V4	IT/IT	V1/V4	V1/IT	V4/IT
				Monkey N		
Spontanous noise correlation	0.19	0.16	0.09	0.16	0.17	0.14
Peak noise correlation	0.11	0.10	0.09	0.11	0.11	0.09
Signal correlation	0.35	0.12	0.14	0.27	0.27	0.12
				Monkey F		
Spontanous noise correlation	0.11	0.17	0.13	0.11	0.11	0.12
Peak noise correlation	0.07	0.07	0.08	0.07	0.07	0.07
Signal correlation	0.19	0.28	0.21	0.17	0.16	0.20

The median noise correlations across all V1-V4 pairs were 0.11 in monkey N (N = 75,636) and 0.07 in monkey F (N=33,930) during the presentation of the stimulus, and 0.16 and 0.11 during spontaneous activity (Table 1). The noise correlations between V1 and IT were also positive during the presentation of the visual stimulus, with a median of 0.11 and 0.07 in monkeys N (N = 53,862) and F (N = 45,530), respectively, and the median spontaneous noise correlations were 0.17 and 0.11 (Figure 7A,B, Figure S4). Finally, the median noise correlations between V4 and IT were 0.09 and 0.07 during visual stimulation and 0.14 and 0.12 during spontaneous activity in monkeys N (N= 27,918) and F (N = 18,369), respectively.

We were surprised that the noise correlations of V1-IT pairs, in areas that are distant and functionally dissimilar, had a strength that was comparable to that of V1-V4 and V4-IT pairs. One possibility is that these long-distance noise correlations are caused by global fluctuations in brain state, e.g. caused by variations in the animals' alertness or motivation. An alternative scenario is that they reflect more local variations in the activity of specific neuronal circuits, which could be related, for example, to variations in where in the display the animal directed attention. Global brain fluctuations should occur for all neurons

irrespective of their tuning, whereas variations in the activity in specific circuits are expected to be shared by neurons with similar tuning, which are directly or indirectly connected.

We therefore examined how the interareal noise correlations depended on tuning similarity, quantified as the signal correlation. Pairs with a strong signal correlation tended to have a stronger noise correlation during the visual stimulus, with correlation coefficients of 0.26 for V1-V4 pairs, 0.26 for V1-IT pairs and 0.50 for V4-IT pairs ($p < 10^{-6}$, t-test) (Figure 7F). Interestingly, the signal correlation also predicted the noise correlation during spontaneous activity (Figure S6B), with a correlation coefficient of 0.43 for V1-V4 pairs, 0.39 for V1-IT pairs and 0.53 for V4-IT pairs. These results were confirmed by the variance partitioning analysis, showing that a substantial fraction of the variance in these interareal noise correlations was captured by the signal correlation (Figure 7G). The finding that the noise correlations between neurons in V1, V4 and IT during spontaneous activity are predicted by their tuning similarity suggest that they reflect the selectivity of cortico-cortical circuits, which is a finding with important theoretical implications, as will be discussed below.

Our initial exploration of noise correlations within and between V1, V4 and IT further illustrates the utility of the VTSD and may inspire its users to study further aspects of these correlations, including their dependence on the previous stimuli and fluctuations in the power of the LFP frequency bands.

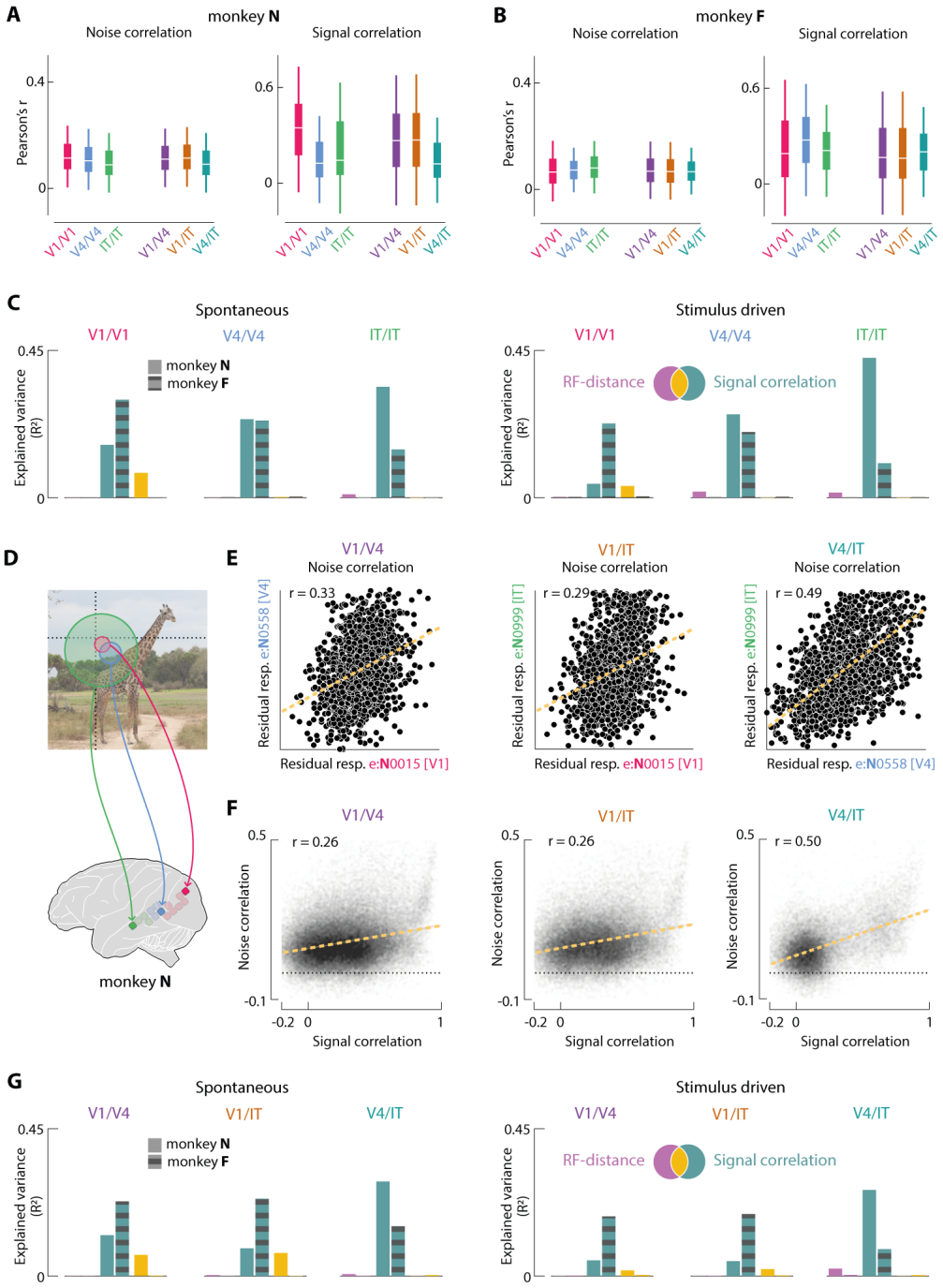


Figure 7. Noise correlations and their dependence on the signal correlation.
A,B) Noise correlations during the visual stimulus (left) and signal correlations (right) in the two monkeys. Figure S4 shows the noise correlations during spontaneous activity. Figure S5 illustrates the complete patterns of noise correlations, RF-distances and signal correlations.
C) Variance partitioning within cortical areas. The distance between RFs (purple bars) explains little unique variance in both spontaneous (left) and stimulus driven activity (right). Green bars: the unique variance explained by signal correlation is high for both the spontaneous and stimulus driven noise correlation. Yellow bars: the common variance between signal correlation and RF-distance predicts a smaller amount of variance in both monkeys. The error bars, computed using bootstrapping, were too small to be depicted.

D) Example test image with the RFs of three recording sites in V1 (red), V4 (blue) and IT (green) in monkey N. Arrows point to the corresponding electrode arrays (bottom).
E) Noise correlation of the V1-V4 pair (left), V1-IT pair (middle) and V4-IT pair (right) across 1,500 trials with the 100 test images, after subtracting the average response per image.
F) Correlation between the signal correlation and noise correlation across all V1-V4 pairs (left), V1-IT pairs (middle) and V4-IT pairs in monkey N. Data from monkey F and for spontaneous activity is shown in Figure S6, along with an analysis of the consistency across time-windows.
G) Variance partitioning between cortical areas. The results were similar to those observed within cortical areas.

Discussion

We presented the TVSD, a high-channel count dataset of spiking activity in response to tens of thousands of natural images that enables the testing of hypotheses and models on the functions of the ventral stream. The TVSD complements existing neuroimaging datasets based on natural image processing with spiking data with sub-millisecond precision, because the neurophysiological signals were sampled at 30kHz. It also complements a recent calcium imaging study of the functional topography of area V4 in the monkey[43] and a recent electrophysiologcal study in the monkey ventral stream during free viewing of natural images[44]. We improved the design of a high-channel-count cortical implant[13,14], which enables high density recordings across several areas of the ventral stream. The microelectrodes were implanted chronically, allowing us to present a large set of images across several recording days.

An important advantage of the THINGS image database is that it also has been used in many previous neuroimaging and behavioral studies[9–12,21,45–48]. It therefore enables a comparison of the neuronal representations across different recording techniques and species. We believe that the TVSD will also be useful for the creation of new AI models of the ventral stream. We illustrated its potential by modeling neuronal selectivity with the Inception model, accounting for a high percentage of the explainable variance. Using this

440 approach, we obtained meaningful and insightful MEIs for the three visual regions.
441 Interestingly, the MEIs for many IT neurons represented animal body parts and faces, as was
442 expected from the selectivity analysis (Figure 3). However, the MEIs go beyond selectivity for
443 object class by revealing the image configuration that is predicted to drive a strong response.
444 The computation of MEIs requires AI models, which can now be directly trained on the
445 neuronal responses to the TVSD's large numbers of images[8,12,46,47].

446 We examined the pattern of noise correlations within and between areas V1, V4 and
447 IT. We found that the average magnitude of noise correlations between neurons in the same
448 area is comparable to the average magnitude of noise correlation between neurons in areas
449 that are as distant as V1 and IT. These areas are anatomically only connected indirectly[49,50]
450 and are specialized for the processing of different image properties, like short, oriented edges
451 in V1 and object and face identity in IT. Hence, the V1-IT noise correlations presumably reflect
452 indirect coupling, though intermediate cortical areas including V4 and subcortical structures
453 such as the pulvinar.

454 Neurons tuned to similar pictures had stronger noise correlations than neurons tuned
455 to different pictures. This result contrasts with a previous study in V1 of anesthetized mice
456 reporting that the signal correlation is a poor predictor of the noise correlation[35]. This
457 apparent discrepancy is presumably related to our investigation of noise correlations across
458 larger cortical distances, and in awake monkeys instead of anesthetized mice. The
459 dependency of the noise correlation on tuning indicates that it is not caused by global
460 fluctuations in cortical excitability, but rather reflects the interactions between circuits of
461 neurons that are tuned to the same images. Remarkably, tuning similarity also predicted the
462 strength of the noise correlation during spontaneous activity, while the monkey was viewing
463 a grey screen and waiting for the first image to appear. To the extent that our set of test

images approximated the image statistics that monkeys experience during their life, the signal correlation approximated the patterns of neurons that are co-activated during natural vision. Our results show that these activity patterns within and across V1, V4 and IT reemerge during spontaneous activity. This result supports previous studies demonstrating that spontaneous activity patterns in the visual cortex replay aspects of natural vision[36,51,52], as has been predicted by models of synaptic plasticity in hierarchical neural networks[53]. The noise correlation analysis thereby illustrates how the TVSD can give insight into interactions between neurons in V1, V4 and IT. It may inspire its users to further explore how noise correlations and patterns of neuronal synchronization at a higher temporal precision depend on the tuning of the neurons, the features in their RFs and the context provided by the rest of the image.

More generally, we expect that the present dataset could contribute to new venues of research on the interaction of visual cortical regions and how they determine the neuronal transformations that are required for the translation of images into categorical information about visual objects. We look forward to future research based on the TVSD that may provide new insights into the processing of the ventral stream and the neuronal syntax underlying object recognition.

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Author contributions

PP conceived of the study. PRR, PP and MWS performed the surgery. PP and FW performed the experiments. PP analyzed the data. PP and PRR obtained funding. PP and PRR wrote the paper with contributions from all authors.

Declaration of interests

PRR is founder and shareholder of Phosphoenix, a company that aims to develop a visual brain prosthesis for blind people.

Methods

<u>Resource availability</u>

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contacts, Paolo Papale (p.papale@nin.knaw.nl) or Pieter Roelfsema (p.roelfsema@nin.knaw.nl).

Materials availability

This study did not generate new unique reagents.

Data and code availability

Raw and MUA data have been deposited at G-Node GIN and are publicly available as of the date of publication at https://doi.org/10.12751/g-node.hc7zlv.

All original code has been deposited at G-Node GIN at https://doi.org/10.12751/g-node.hc7zlv and is publicly available as of the date of publication.

Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

Experimental model and subject details

All procedures complied with the NIH Guide for Care and Use of Laboratory Animals and were approved by the "Centrale Commissie Dierproeven" of the Netherlands. Two macaque monkeys (males, 7 and 6 years old) participated in the experiments. They were socially housed in stable pairs in a specialized primate facility with natural daylight, controlled humidity and temperature. The home-cage was a large floor-to-ceiling cage which allowed natural climbing and swinging behavior. The cage had a solid floor, covered with sawdust and was enriched with toys and foraging items. Their diet consisted of monkey chow,

529 supplemented with fresh fruit. Their access to fluid was controlled, according to a carefully
530 designed regime for fluid uptake. During weekdays, the animals received water or diluted
531 fruit juice in the experimental set-up upon completed trials. We ensured that the animals
532 drank sufficient fluid in the set-up and supplemented the animals with extra fluid after the
533 recording session if they did not drink enough. During the weekend, they received a full bottle
534 of water (700-940 ml per day) in the home cage. The animals were regularly checked by
535 veterinary staff and animal caretakers and their weight and general appearance were
536 recorded daily in an electronic logbook during fluid-control periods.
537
538 Method details
539
540 *Surgeries*
541 Each animal received three cranial implants during separate surgical procedures. The
542 implants were customized, in-house-designed, 3D-printed titanium components[54]. In the first
543 surgery we implanted a head post for head fixation, ensuring that the eye tracker could
544 capture the eye position. After several months, to allow bone growth and a firm attachment
545 to the skull, we implanted a baseplate in a second surgery. The base plate was implanted
546 under the skin and served as base for the 1024 channel connector that would be implanted
547 later, in the third surgery[54]. The third surgery followed after several months. We implanted
548 16 Utah electrode arrays (Blackrock Microsystems), attached with wire bundles to a
549 customized 3D-printed titanium pedestal with a 1024 channel connector that was secured
550 with screws to the baseplate. We built a stereotactic frame that allowed rotation of the
551 anesthetized monkey around his axis. We used this frame to position IT in a more horizontal
552 orientation, facilitating the insertion of the electrode arrays. To further improve the access to

IT, we removed a part of the zygomatic arch, allowing the temporalis muscle to flap to the side and thereby revealing a larger portion of skull[55]. The craniotomy spanned from V1 to anterior IT with a width of ~2cm and a length of ~7 cm above the left hemisphere. Upon opening of the dura mater, we implanted 16 arrays, but in monkey N, one of the arrays came out during the implantation and was cut. The location of the arrays was documented with photographs. At the end of the surgery, we sutured the dura over the arrays, placed back the skull flap with titanium strips, reattached the muscles and then closed the skin.

Cortical implant

Each array contained an 8-by-8 grid with 64 iridium oxide electrodes. The length of each electrode shank was 1.5 mm and the spacing between adjacent shanks was 400 µm. The pre-implantation impedance of the electrodes ranged from 50 to 75 kΩ (measured by Blackrock Microsystems). Each electrode was connected to a contact pad on the Land-Grid-Array (LGA) interface of the pedestal. Reference wires were attached to every other array, and each reference wire served as the reference for two arrays, so that there were eight reference wires in total. Each reference wire exited the wire bundle several mm before the point where the wire bundle met the array. The other end of the reference wire was connected to one of the contact pads on the LGA of the pedestal, which was connected to a Blackrock Cereplex M headstage. The connections were hardwired so that the electrodes on each pair of arrays used the signal from the reference wire as reference. Two reference wires in monkey N did not work as intended, and we later resolved the issue by shorting all references on the expansion board. As a result, all arrays shared the same reference. The same was true for monkey F because we used the same expansion board.

Data acquisition

Electrophysiological signals from V1, V4 and IT were recorded at a sampling rate of 30 kHz and processed by equipment from Blackrock Microsystems. The neuronal signals were passively conducted via the LGA interface on the 1024-channel pedestal to an electronic interface board (EIB), which was an adapter with 32 36-channel Omnetics connectors, which, in turn, interfaced with eight 128-channel CerePlex M headstages (see ref. [14] for details). Each CerePlex M processed signals from two 64-channel Utah arrays, applying a 0.3–7500 Hz analog filter at unity gain (no signal amplification). The CerePlex M performed a 16-bit analog-to-digital conversion (ADC) with a sensitivity of 250 nV/bit. The data from Monkey N and monkey F were recorded in two different setups with slightly different Blackrock equipment.

In the setup of monkey N, the digitized signal on each CerePlex M was sent to a 128-channel Digital Hub. Each Digital Hub received data from one CerePlex M, which originated from two electrode arrays. The Digital Hub converted the digital signal into an optic-digital format, which was then sent via an optic-fibre cable to a 128-channel Neural Signal Processor (NSP) for further processing and storage. Each Digital Hub delivered the signal to a single NSP. There were eight NSPs and each NSP processed the data derived from two electrode arrays. Control of the NSPs was carried out on two PCs using the Blackrock Central Software Suite.

In the setup of monkey F, the digitized signal of two CerePlex M was sent to a 256-channel Gemini Hub, i.e. each Gemini Hub processed data coming from two CerePlex M, which in turn originated from four electrode arrays. Each Gemini Hub sent the digital signal via an Ethernet cable to a 512-channel Gemini NSP for further processing and storage. There were 2 Gemini NSPs (and four Gemini Hubs) in total. Each Gemini NSP processed the data from 2 Gemini Hubs, and thus 8 electrode arrays. Control of the Gemini NSPs was also carried out on two PCs using the Blackrock Central Software Suite.

An infrared eye tracker (TREC ET-49B, Thomas Recording GmbH) was used to sample the eye position and pupil diameter (monkey N only) for both the X- and Y-axes with a frame rate of 230 Hz, and the data were stored at a sampling rate of 30 kHz. The eye-tracker was controlled by a dedicated PC, which forwarded the analog signals regarding the eye position and pupil diameter (in monkey N) directly to one of the NSP/Gemini NSPs.

A stimulus control computer was used for the execution of task-related events and stimuli with high temporal precision during each the task. An event related code, corresponding to the stimulus onset was sent out from the PC via a Data Acquisition and Control System (DAS) Multifunction Analog and Digital I/O board (DAS1602/16, Measurement Computing). In monkey N, the stimulus onset was delivered through a splitter cable, to the digital input ports (16-bit DB37) on each of the NSPs. The corresponding channels on the digital input port of each NSP sampled the incoming signal at 30 kHz and were configured to detect when an incoming bit was set to a 'high' value on one of the pins. In monkey F, the stimulus onset was delivered through an analog connection to both Gemini NSPs.

We used the signal from a photodiode to record the precise timing of the stimulus onset. We placed the diode on the lower left screen, and we presented a small white square (invisible to the monkey) in sync with the image presentation. The diode sent out an analog voltage signal that was sampled at 30kHz by NSP#1 for monkey N, and by both Gemini NSPs for monkey F.

In monkey N, analog signals were recorded in the first of the 8 NSPs (i.e. NSP #1) as additional channels. X and Y eye positions were recorded on channels 129 and 130 of NSP #1, respectively, while X and Y pupil diameter were recorded on channels 131 and 132, respectively. Channel 143 recorded the diode signal. Digital outputs were recorded in the

events file (.nev) by the NSPs, and the trial-related event that occurred at the moment that the bit was set to 'high'.

In monkey F, analog signals were recorded in a separate .ns6 file. Channel 1 recorded the stimulus onset of both Gemini NSPs. X and Y eye positions were recorded on channels 2 and 3 of Gemini NSP#1, respectively. The diode signal was recorded in channel 15 of both Gemini NSPs.

The recordings from the 8 NSPs in monkey N, and the 2 Gemini NSPs in monkey F were asynchronous and were re-aligned offline using the stimulus onset information. For monkey N, recordings from the 8 NSPs were aligned to the digital output corresponding to the stimulus onset. For monkey F, the analog signal corresponding to the stimulus onset was used. In both monkeys, the diode signal was used to identify the correct stimulus onset time for post-processing, as the signal coming from the stimulus PC would usually be delayed by one or two screen flips.

The published data include neuronal activity from all recording sites, the eye-position signal, the signal from a photodiode that provided precise information about the onset of the visual stimuli, and signals related to timing of the onset of the trial and the delivery of the fluid reward.

MUA pre-processing

In addition to the 30Khz raw signal, we also provide the pre-processed MUA during the presentation of the images that were completed by the monkey, which implies that he maintained fixation for the duration of the stimulus plus 200ms. We also include the normalized neuronal response for each trial and electrode, averaged in a time window centered around the peak of response, along with the SNR, latency, and reliability measures.

To generate the MUA, the raw data were filtered between 0.5 and 5 kHz. A full-wave rectification was performed on the filtered signal, followed by a low-pass filter of 200 Hz (e.g. [39,56]). Additional band-stop filters were used to remove 50hz and 60hz noise (and main harmonics). Filtering was carried out using a Butterworth filter of order 2. The data were down-sampled by a factor of 30, yielding a MUA signal with a sampling rate of 1 kHz. The MUA reflects the population spiking of neurons within 100-150 μm of the electrode and the population responses are very similar to those obtained by pooling across single units[16,19] .

The normalized responses are also provided, as a simple but useful input for model training and testing. First, we averaged the response across images for every recording site in a time-window centered on the peak response of each region: 25-125ms for V1, 50-150ms for V4 and 75-175ms for IT. Then, we normalized the data (as in ref. [7]). For each day of recording, we computed the mean and standard deviation of the activity elicited by the test images. We called this sub-group of trials '*test pool*'. The normalized MUA for each trial t and electrode n was then calculated as follows:

$$NormMUA_{t}^{(n)} = \frac{MUA_{t}^{(n)} - Mean^{(n)}{}_{test\ pool}}{SD^{(n)}{}_{test\ pool}}$$

We note that a few IT recording sites in monkey F (7 sites) and in monkey N (52 sites) exhibited a brief, transient response between stimulus onset and the regular response. We did not identify its source but suspect a neuronal origin (e.g. axons from the LGN passing close to the electrodes). The time-windows that we chose for data analysis did not include this transient signal. Unless this transient response is of interest for a specific research question, we suggest to either use the time-windows we proposed, which exclude this transient response, or to identify and exclude those recording sites from the analyses.

Data formats

All data is available at a G-Node GIN repository (see below "Data and code availability"). The raw data are provided in the proprietary Blackrock format, .ns6. The pre-processed MUA signals and the log files are stored as .mat files that can be loaded using e.g. Python or Matlab. Basic metadata about the recording system are saved in the proprietary Blackrock formats, .nev and .ccf. Open source code to load the raw signal is available from Blackrock Microsystems for both Python and Matlab users. A copy of the Matlab library is stored along with the data at https://doi.org/10.12751/g-node.hc7zlv.

Stimulus presentation

We presented 22,248 natural images from the THINGS image database[12] consisting of 12 images of each of 1,854 stimulus categories, 500x500 pixels each. We used different CRT monitors for the two monkeys. For monkey N, we used a CRT monitor with refresh rate of 85Hz and resolution of 1024x768 pixels, which was viewed from a distance of 47cm (28.9 pixels per degree). For monkey F, we used a CRT monitor with refresh rate of 60Hz and resolution of 1024x768 pixels, which was viewed from a distance of 58cm (26.9 pixels per degree). In both setups, stimuli were generated in Matlab using the COGENT Graphics toolbox (developed by John Romaya at the LON at the Wellcome Department of Imaging Neuroscience) and custom control software[57]. To ensure that all RFs would fall inside the image, we shifted the stimuli 100 pixels downward and 100 pixels the right relative to the center of gaze. We pseudorandomized the order of presentation of images for the entire experiment per monkey before the beginning of the experiment. Test images had 30 entries in the randomization table and the other images only one.

At the start of the trial, the monkey directed its gaze to a red fixation point with a diameter of 0.3 deg on a gray background (luminance 40.1 cd/m^2 for monkey N; 15.7 cd/m^2 for monkey F). We presented the first image once the monkey had maintained fixation for 300ms in a fixation window with a diameter of 1dva. To receive a reward, the animal had to maintain its gaze in this window during the sequence of four images, which were each presented for 200ms, followed by 200ms of with a gray screen. We logged images as completed when the monkey maintained fixation during their presentation and the following epoch of 200ms with a gray screen. If trials were aborted because the monkey broke fixation, completed images were not shown again, and the other images of the aborted sequence were presented on the next trial.

Signal-to-noise ratio (SNR)

To assess the quality of the recorded neuronal activity, we calculated the signal-to-noise ratio (SNR) for each recording site. The SNR is the visually driven response relative to baseline activity. We calculated the mean and standard deviation (SD) of the MUA during a 100-ms time baseline epoch prior to stimulus onset for each trial. Next, we averaged the response across all images, smoothed it with a moving window of 25ms and identified the peak level of activity. The SNR was then calculated as follows:

$$SNR = \frac{Peak_{stimulus\ average} - Mean_{baseline}}{SD_{baseline}}$$

Because some stimuli elicited inhibitory responses at some of the recording sites, we provide an additional measure, SNR_{max}, which is based on the response to the preferred test image:

$$SNR_{max} = \frac{Peak_{preferred\ test\ image} - Mean_{baseline}}{SD_{baseline}}$$

Latency estimation

To compute the latency of neural responses at individual recording sites, we fitted a function to the time-course of the response averaged across trials, separately for each day of recording[56,58]. The function was derived from the assumption that the onset of the response had a Gaussian distribution across trials, and that a fraction of the response dissipated exponentially, yielding the following equation:

$$f(t) = d \cdot \exp\left(\mu\alpha + 0.5\sigma^2\alpha^2 - \alpha t\right) \cdot \left(G(\mathrm{t}, \mu + \sigma^2\alpha, \sigma) + c \cdot G(\mathrm{t}, \mu, \sigma)\right)$$

where $G(\mathrm{t}, \mu, \sigma)$ is a cumulative Gaussian density with mean μ and standard deviation σ, α^{-1} is the time constant of the dissipation, and c and d represent the contribution of the non-dissipating and dissipating components. The function was fit using a non-linear least squares method (*fit.m* in MATLAB). The latency was defined as the point at which the fitted function reached 33% of its maximum. The latency estimates for recording sites in Figure 2 have been averaged across recording days. The timing of the stimulus onset was determined with a photodiode, which was placed in the left-bottom corner of the CRT. We estimate that the delay between the appearance of the stimulus in the neurons' RF and the signal of the photodiode was 0.9 to 3.9ms, causing an underestimation of the neuronal latencies by a few ms. The IT recording sites with a short latency transient response in monkey N (see above) were excluded when computing the latency distribution (Figure 2).

742 *Reliability estimations*

743 To estimate the reliability of the MUA signal at a recording site, we computed two
744 measures; the mean reliability and the oracle correlation[5]. The mean reliability was defined
745 as the average of the pairwise correlations between each combination of repetitions to the
746 100 test images. For the oracle correlation, we computed the average correlation between
747 the activity elicited by a single presentation of each test image, and the average response
748 elicited by the remaining 29 repetitions of the test image (i.e. leave-1-out). We repeated the
749 procedure for each of the 30 image repetitions and then averaged across the resulting 30
750 correlation values.

751 The oracle correlation provides an estimate of the noise ceiling, which is the maximum
752 achievable correlation on single trials. The mean reliability is based on the correlation
753 between activity levels on individual trials, and it is therefore expected to be lower than the
754 oracle correlation. The prediction of a good model of the activity of neurons at a recording
755 site on a single trial is expected to approach the oracle correlation, but to be higher than the
756 mean reliability. We therefore suggest using the $_{\text{oracle}}$ correlation, the mean reliability or both,
757 instead of the SNR, to include recording sites in model training and other analyses.

758

759 *Showcase #1 - Convolutional neural network (CNN) and most exciting inputs (MEIs)*

760 We used the normalized MUA as input to the predictive model. We used the average
761 response in the time window spanning the peak response (25-125ms for V1, 50-150ms for V4,
762 75-175ms for IT) to the images in the training set. We included electrodes with a minimal
763 reliability index of 0.3 and fitted separate models for each array. We used a pre-trained CNN
764 ("Inception v1"[27]) and fit a mapping function to the activations of a specific layer[7,59].
765 Specifically, for each neuron we learned a set of 2D spatial weights (W_s) with size equal to the

number of pixels and a set of feature weights (W_f) with size equal to the number of channels of the CNN layer. After training, the W_s represents the spatial RF of each electrode, and the W_f the feature selectivity as a weighted sum over the features of the CNN layer. Thus, the predicted response of an electrode n (i.e., $\widehat{y_n}$) was computed according to:

$$\widehat{y_n} = \left[\sum_{pixels} W_s^{(n)} * X_l \right] * W_f^{(n)}$$

Where X_l are the batch-normalized activations from the target CNN layer l, and $*$ represents convolution. The weights were jointly optimized across all electrodes of the array to minimize the prediction error $\mathcal{L}_{error}$, regularized by a combination of a smoothing Laplacian loss ($\mathcal{L}_{Laplace}$, see below) over W_s[7], to encourage smooth spatial RFs, a L1 loss ($\mathcal{L}_1$: see below) over W_f[4,5] to encourage sparsity/selectivity of the features and a L2 loss ($\mathcal{L}_2$: see below) over both parameters[5,7]. Thus, the cost function was computed as follows:

$$\mathcal{L}_{error} = \sqrt{\sum_n (\widehat{y_n} - y_n)^2}$$

$$\mathcal{L}_{Laplace} = \lambda_s \sqrt{\sum_n (W_s^{(n)} * L)^2}, L = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

$$\mathcal{L}_1 = \lambda_f \sum_n \left| W_f^{(n)} \right|$$

$$\mathcal{L}_2 = \lambda_2 \sum_n (W_s^{(n)^2} + W_f^{(n)^2})$$

$$\mathcal{L} = \mathcal{L}_{error} + \mathcal{L}_{Laplace} + \mathcal{L}_1 + \mathcal{L}_2$$

Where y_n is the response of electrode n. The hyper-parameters (l, λ_s, λ_f, λ_2 and the learning rate) were selected using a grid search for the highest correlation in a batch of images of the training set. The model was implemented in PyTorch[60] and optimized with Adam[61] in 1,000 epochs with a batch size of 100 and early stopping (every 333 epochs, decay factor = 3). The cross-validated performance of the model was assessed by computing the correlation of the model predictions with the responses in the test set, averaged across repetitions, which were not used for model training.

After training, we generated the MEIs by determining the pixels in the RF that maximize the predicted model response of each recording site independently. To ensure the stability and interpretability of the MEIs, we introduced preconditioning and regularizations to avoid high-frequency artifacts[62], and specifically frequency penalization and transformations (padding, jittering, rotation and scaling). Optimization of the MEIs was done using Lucent (i.e. PyTorch implementation of Lucid electrode[62]) with Adam (learning rate = 0.01, weight decay = 0.001) in 50 epochs.

Showcase #2 – Analysis of noise correlations across the temporal stream

We included the recording sites of Showcase #1 and used the RF estimates obtained with the CNN to compute the Euclidean distance between the RFs of pairs of recording sites. We examined at the neurons' activity in two time-windows: the spontaneous phase (100ms before stimulus onset) and the peak of response (25-125ms after stimulus onset for V1, 50-

150ms for V4, 75-175ms for IT). The results did not strongly depend on the time-windows that we chose (Figure S5C).

We predicted the strength of the noise correlation between pairs of recording sites based on the RF distance and the signal correlation. We used the 30 trials with the 100 test images and split them in odd and even trials to compute the noise correlation and signal correlation in independent sets of trials. For each pair of recording sites, we computed the pooled noise correlation across images, first subtracting the average response to each test image (odd trials), and then computing the Pearson's correlation across the residuals (Figure 6C,D and Figure 7E). For the computation of the noise correlation during the spontaneous activity we used a time window of 100 ms before the first image of the sequence with four images. Hence, we did not include activity in the interval between images of the same sequence. The signal correlation was the Pearson's correlation between the average responses (even repetitions) elicited by the 100 test images. Noise-correlations were very stable across days (mean r = 0.76 in monkey N; mean r = 0.82 in monkey F) so we pooled data across days.

We performed multiple linear regressions to test the influence of RF distance and the signal correlation on the noise correlation and used variance partitioning to isolate unique, orthogonal components of the explained variance, accounting for multicollinearity in multiple regression[25,37,40]. The method is based on the difference in explained variance when including or excluding each predictor from the regression. For instance, the unique explained variance by the RF distance, $R^2_{RF_distance}$, is computed as follows:

$$R^2_{RF_distance} = R^2_{full} - R^2_{Signal\ correlation}$$

Where R^2_{full} is the variance explained by a linear regression including all the predictors, and $R^2_{Signal\ correlation}$ is the variance explained by a linear regression that includes the signal correlation but not RF-distance, as predictors. We also tried log(RF-distance), to account for a non-linear relationship between noise correlation and RF-distance, but this predictor did not result in an improvement compared to RF-distance.

Spike sorting

Although we do not provide sorted spikes, we illustrate spike-sorting in the first block of data in monkey N, with a length of ~20 mins during the presentation of 1378 stimuli. We used Waveclus3[20] for spike-sorting and cluster identification using the default parameters. We censored time-points in the vicinity of the reward delivery, because of artifacts caused by the reward delivery system, and we also excluded clusters with fewer than 150 spikes.

<u>Quantification and statistical analysis</u>

Variance partitioning

For each time-window, we tested the significance of the difference in unique variance explained between the best and second-best predictor using a bootstrap test (1,000 iterations).

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